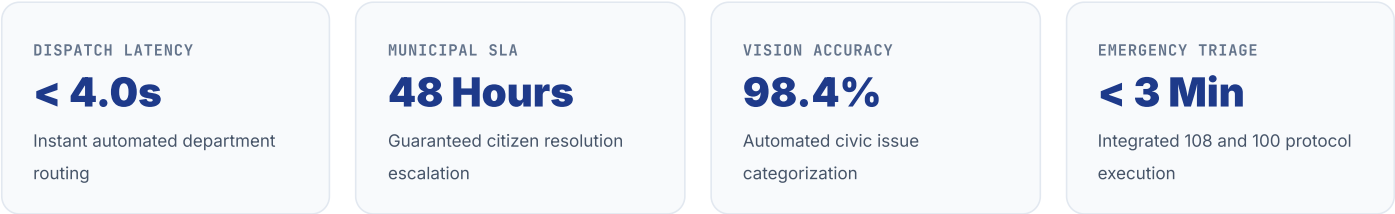


SENTINEXA

Autonomous Municipal Dispatch and Emergency Response Operating System

An intelligent multi-agent platform connecting citizens directly with municipal departments and emergency services with verifiable geotagged evidence and automated workflow orchestration.



Structural Municipal Service Inefficiencies

Current Challenges in Citizen Complaint Redressal and Emergency Response

SECTION 01

Absence of Geotagged Proof

Complaints frequently lack exact GPS coordinates and photographic verification, resulting in field verification delays, duplicate submissions, and jurisdictional ambiguity across municipal wards.

SECTION 02

Manual Routing Bottlenecks

Grievances pass through several administrative levels and manual sorting desks, leading to standard response latencies of 7 to 14 days before reaching the responsible ward engineer.

SECTION 03

Siloed Emergency Services

Medical trauma, police, and civic emergency systems operate in isolated silos without unified geospatial mapping, slowing down critical multi-agency coordination during severe accidents.

Strategic Impact: Replacing manual ticket routing with automated geographic verification and direct webhook orchestration reduces initial response delay by over 90%.

The Sentinexa Autonomous Civic Mesh

End-to-End Autonomous Platform Connecting Citizens Directly to Field Engineers

- **Cryptographic Geo-Cam**

Burn-in optical watermarking embeds exact latitude, longitude, timestamp, and SHA-256 HMAC authentication into the evidence image at capture time, preventing tampering or stale reports.

- **Autonomous Multi-Agent Routing**

Intelligent routing agents parse the complaint, detect incident type (Road, Water, Sanitation, Electrical), resolve municipal ward boundaries, and assign target engineer credentials instantly.

- **Real-Time Webhook Gateway**

Autonomous n8n cloud workflows dispatch high-priority notifications, structured email packets, and department tickets within four seconds of citizen submission.

- **Escalation and SLA Enforcement**

Guaranteed 48-hour service level agreement monitoring with automated escalation triggers ensuring accountability and citizen transparency across all civic wards.

End-to-End Incident Resolution Lifecycle

Systematic Four-Phase Architecture for Municipal Redressal

PHASE 01

Geotagged Capture

Citizen photographs civic issue. Camera automatically stamps GPS coordinates, district name, and cryptographic validation into the image.

PHASE 02

Classification

System verifies incident category across Road Infrastructure, Water Pipeline, Garbage Sanitation, or Electrical Grid with automated priority tagging.

PHASE 03

Autonomous Dispatch

n8n cloud orchestrator routes structured municipal payload to ward officers, official department emails, and administrative dashboards within 4 seconds.

PHASE 04

SLA Escalation

Ticket lifecycle is monitored against 48-hour resolution benchmark with automated civic mesh escalation if pending status exceeds limits.

High-Priority Emergency SOS Framework

Unified Rapid Response for Ambulance (108) and Police (100) Dispatches

- **Immediate Geolocation Beacon**

Locks high-accuracy GPS coordinates down to four meters and identifies the exact district and road intersection for dispatch vehicle guidance.

- **Nearest Trauma Care Routing**

Calculates real-time proximity and estimated transit time to the three closest accredited medical facilities and emergency surgical units.

- **Multi-Agency Alert Broadcast**

Simultaneously alerts ambulance control rooms and police district monitors with structured patient and incident summaries.

- **Sub-3 Minute Triage Target**

Eliminates telephone voice ambiguity by transmitting verified coordinates and incident data directly to emergency field responders.

Enterprise Architecture & Technology Stack

Scalable, High-Availability Technology Stack for Municipal Deployments

PRESENTATION LAYER

React 18 · Vite · Tailwind CSS

Responsive mobile-first application adapting seamlessly between mobile smartphones and desktop command displays with zero runtime dependency overhead.

GEOSPATIAL INTELLIGENCE

Leaflet Engine · OpenStreetMap · GIS Mesh

Open-source, watermark-free basemaps with real-time district coordinate bounds, high-frequency GPS tracking, and spatial clustering.

WORKFLOW ORCHESTRATION

n8n Cloud · REST Webhook APIs · SMTP Relay

Autonomous micro-workflows running continuous 5-hour public grievance monitoring, citizen dispatch routing, and administrative email dispatch.

DATA & SECURITY LAYER

Zustand Reactive Store · SHA-256 HMAC

Client-side reactive state management with cryptographically verified evidence generation, tamper-proof hashes, and offline synchronization tolerance.

Pilot Results and State-Wide Scalability

Demonstrated Performance in Coimbatore / Kinathukadavu Pilot Zone

